

Question	Answer
What will be the output of the following Python code? <code>print("Hello {0!r} and {0!s}".format('foo', 'bin'))</code>	Hello 'foo' and foo
Which of these about a set is not true?	Immutable data type
Which of the following is not the correct syntax for creating a set?	<code>set([[1,2],[3,4]])</code>
Which of the following statements is used to create an empty set?	<code>set()</code>
If <code>a={5,6,7,8}</code> , which of the following statements is false?	<code>a[2]=45</code>
Which of these about a frozenset is not true?	Mutable data type
What will be the output of the following Python code? <code>s=set() type(s)</code>	
Set makes use of _____ Dictionary makes use of _____	keys, key values
Which of the following lines of code will result in an error?	<code>s={san}</code>
What will be the output of the following Python code? <code>s={2, 5, 6, 6, 7}</code> s	<code>{2, 5, 6, 7}</code>
Which of the following functions cannot be used on heterogeneous sets?	sum
Which of the following functions will return the symmetric difference between two sets, x and y?	x & y
The difference between the functions discard and remove is that:	Remove throws an error if the specified element is not present in the set whereas discard does not throw an error in case of absence of the specified element
Which of the following is a Python tuple?	(1, 2, 3)
Suppose <code>t = (1, 2, 4, 3)</code> , which of the following is incorrect?	<code>t[3] = 45</code>
What is the data type of (1)?	Integer
Is the following Python code valid? <code>>>> a,b=1,2,3</code>	No, too many values to unpack

Is the following Python code valid? >>> a=(1,2,3) >>> b=a.update(4,)	No because tuples are immutable
Is Python case sensitive when dealing with identifiers?	yes
What is the maximum possible length of an identifier?	none of the mentioned
Which of the following is invalid?	none of the mentioned
Which of the following is an invalid variable?	1st_string
Why are local variable names beginning with an underscore discouraged?	they are used to indicate a private variables of a class
Which of the following is not a keyword?	eval
All keywords in Python are in _____	None of the mentioned
Which of the following is an invalid statement?	a b c = 1000 2000 3000
Which of the following cannot be a variable?	in
Which is the correct operator for power(xy)?	X**y
What is the output of this expression, 3*1**3?	3
Which one of the following has the highest precedence in the expression?	Parentheses

Question	Correct Answer
Which of the following is NOT supported by MySQL	Stored Procedures
Commit command is used for	To save the current transaction
Which of the following is available in MySQL:	CREATE DATABASE
The REAL type is synonym for _____	DOUBLE
mysql_init() returns a _____	pointer to a structure
How do the STRICT_ALL_TABLES and STRICT_TRANS_TABLES mode values deal with bad data?	reject them
How are identifiers quoted in MySQL?	backticks
The default case sensitivity of database and table names depends on _____	Operating system of machine
Which of the following is case sensitive?	Trigger name

Which clause can be used to sort string values according to a specific collation?	COLLATE
Which statement is used to select a default database?	USE
Which keyword is used to create a database?	CREATE
Which file is created by the server to store the database attributes?	db.opt
What is the clause that filters JOIN results called?	WHERE
What are CROSS JOIN and JOIN are similar to?	INNER JOIN
The stored-routine privileges are contained in _____	procs_priv
The statement used to obtain access privileges to an account is _____	GRANT
The operators used when a subquery returns multiple rows to be evaluated in comparison to the outer query are _____	IN and NOT IN
The ALL subquery performs the operation _____	column
The keyword used with UNION that retains duplicate rows is _____	ALL
The clause that is used to display information that matches a given pattern is _____	LIKE
What does mysql_init() return?	pointer to a structure
Which system variable when set to 1, makes all temporary tables to get stored on disk rather than in memory?	big_tables
Which string function returns the index of the first occurrence of substring?	INSTR()
Which language is the core for APIs and communication in MySQL?	C
Which statement upgrades the database directory name encoding?	ALTER DATABASE
Which statement can produce the same output as 'SHOW COLUMNS'?	DESCRIBE
Select odd one out?	Between
What will be the output of the following MySQL statement?	All employees details

SELECT * FROM employee WHERE start_date BETWEEN '2007-01-01' AND '2008-01-01';	between 2007 and 2008
Fill the blanks with suitable options? BETWEEN _____ AND _____	Lower and upper limit

Correct Option	Correct Answer
Character data can be stored as _____	Either Fixed or Variable length string
Which declaration represents that "character data will consume the same number of bytes as declared and is right padded"?	Char
Which declaration doesn't use the same number of bytes and consumption of bytes depends on the input data?	Varchar
The maximum length of the varchar columns is _____	Upto 65, 535 bytes
Which of the following clauses is used to display information that match a given pattern?	LIKE
What column names are displayed when this SQL command is executed? SHOW COLUMNS FROM tbl_name LIKE '%name';	suffixed with 'name'
The special database that always exists after setting up MySQL in a computer is _____	information_schema
The NULL value also means _____	unknown value
Which clause is used to sort query elements?	ORDER BY
Suppose you want to select a database named 'sampledb' as the default database. Which of the following commands do you use?	USE sampledb
To see the table structure, which of the following SQL commands is issued?	DESCRIBE tbl_name
On UNIX, statements entered in 'MySQL' are saved in which file?	.mysql_history
The system variable in MySQL server that enables to configure the SQL mode is _____	sql_mode
How do the STRICT_ALL_TABLES and STRICT_TRANS_TABLES mode values deal with bad data?	reject them

Which type of database management system is MySQL?	Relational
The MySQL server used in its client/server architecture is _____	mysqld
In MySQL databases, the structure representing the organizational views of the entire databases is _____	Schema
What will happen if the data being loaded into a text column exceeds the maximum size of that type?	Data will be truncated
What will be the storage pattern for "float(4,2)" in Mysql?	Total of four digits, two to the left of decimal and two to the right of decimal
Which among the following are the correct representation of "float(4,2)"?	Both 24.33 and 124.4
What is the default format for "Timestamp" data type?	YYYY-MM-DD HH:MI:SS
Is "Datetime" and "Timestamp" are same data type?	Yes
The datatype for single precision floating point number is _____	FLOAT
How many of the following attributes are allowed only for numeric types? UNSIGNED, ZEROFILL, CHARACTER SET, COLLATE	2
The datatype SMALLINT stores _____	16 bit
What is the command to see the warning messages?	SHOW WARNINGS
The SQL mode to check for divide by zero error is _____	ERROR_FOR_DIVISION _BY_ZERO
Which keyword suppresses errors?	IGNORE
Which mode prevents entry of the 'zero' date value in strict mode?	NO_ZERO_DATE
Which data type is more suitable for storing "small notes" in Mysql?	Mediumtext

Correct Option	Correct Answer
_____ represents an entity in the real world with its identity and behaviour.	An object
What is the output of the following code? <code>class test: def __init__(self,a): self.a=a def display(self): print(self.a) obj=test() obj.display()</code>	Error as one argument is required while creating the object
Is the following piece of code correct? <code>>>> class A: def __init__(self,b): self.b=b def display(self): print(self.b) >>> obj=A("Hello") >>> del obj</code>	True
Predict the output of following python programs: <code>r = lambda q: q * 2 s = lambda q: q * 3 x = 2 x = r(x) x = s(x) x = r(x) print x</code>	24
Which of these is false about recursion?	Recursive functions run faster than non-recursive function
Fill in the line of code for calculating the factorial of a number. <code>def fact(num): if num == 0: return 1 else: return _____</code>	<code>num*fact(num-1)</code>
What is tail recursion?	A function where the recursive call is the last thing executed by the function
Observe the following piece of code? <code>def a(n): if n == 0: return 0 else: return n*a(n - 1) def b(n, tot): if n == 0: return tot else: return b(n-2, tot-2)</code>	<code>b()</code> is tail recursive but <code>a()</code> isn't
What is the output of the code shown below? <code>def f1(): x=100 print(x) x+=1 f1()</code>	100
What is the output of the code shown below? <code>def f1(x): global x x+=1 print(x) f1(15) print("hello")</code>	error
Which of the following is correct about Python?	All of the above
Which of the following statement causes the loop to skip the remainder of its body and immediately retest its condition prior to reiterating?	continue
Which of the following function checks in a string that all characters are alphanumeric?	<code>isalnum()</code>
What is the output of 3 in <code>[1, 2, 3]</code> ?	true

What is the output of the code shown below? <code>def f(p, q, r): global s p = 10 q = 20 r = 30 s = 40 print(p,q,r,s) p,q,r,s = 1,2,3,4 f(5,10,15)</code>	10 20 30 40
Correct Name of caller function	<code>func([param])</code>
What is the output of the following code? <code>a=[13,56,17] a.append([87]) a.extend([45,67]) print(a)</code>	[13, 56, 17, [87], 45, 67].
What is the output of the following code? <code>lst=[[1,2],[3,4]] print(sum(lst,[]))</code>	[1,2,3,4].
_____ exceptions are raised as a result of an error in opening a particular file.	IOError
What is the output of the code shown below? <code>#generator def f(x): yield x+1 g=f(8) print(next(g))</code>	9

Correct Option	Correct Answer
1. Which of these is not a core datatype?	Class
Given a function that does not return any value, What value is thrown by default when executed in shell.	none
What is the return value of <code>trunc()</code> ?	int
1. What is the output of the following? <code>print('{0:.2}'.format(1/3))</code>	0.33
6. What is the output of the following code? <code>word1="Apple" word2="Apple" list1=[1,2,3] list2=[1,2,3] print(word1 is word2) print(list1 is list2)</code>	True False
6. What is the output of the following code? <code>l=[] def convert(b): if(b==0): return l dig=b%2 l.append(dig) convert(b//2) convert(6) l.reverse() for i in l: print(i,end="")</code>	110
12. What happens if the base condition isn't defined in recursive programs?	Program gets into an infinite loop
Which function overloads the + operator?	<code>__add__()</code>
Let A and B be objects of class Foo. Which functions are called when <code>print(A + B)</code> is executed?	<code>__add__()</code> , <code>__str__()</code>
The snippet of code shown below results in an error. State whether true or false. <code>c=re.compile(r'(d+)([A-Z]+)([a-z]+)')</code> <code>c.groupindex</code>	false

Is the output of the function <code>abs()</code> the same as that of the function <code>math.fabs()</code> ?	sometimes
What is the output of <code>print(math.factorial(4.5))</code> ?	error
Which of the following functions does not necessarily accept only iterables as arguments?	<code>chr()</code>
Program code making use of a given module is called a _____ of the module.	Client
What is the output of below program? <code>def cube(x): return x * x * x x = cube(3) print x</code>	27
What is the output of this program? <code>import sys sys.stdout.write(' Hello ') sys.stdout.write('Python ')</code>	Hello Python
What is the output of the following? <code>print('{: #}'.format(1112223334))</code>	1112223334
What is the output of the following? <code>x = "abcdef" i = "a" while i in x[1:]: print(i, end = " ")</code>	no output
What is the output of the following? <code>print('1@a'.isprintable())</code>	true
What is the output of the following piece of code? <code>a = [1, 2, 3, 4, 5] b = lambda x: (b(x[1:]) + x[:1]) if x else [] print(b(a))</code>	[].
Which of the following is not a declaration of the dictionary?	{1,"A",2"B"}
Execute the following in Python shell? <code>>>> a={1:"A",2:"B",3:"C"} >>> a.items()</code>	dict_items([(1, 'A'), (2, 'B'), (3, 'C')])
What is the output of the code shown below if the system date is: 6/19/2017 <code>tday=datetime.date.today() tdelta=datetime.timedelta(days=10) print(tday+tdelta)</code>	2017-06-29
What is the output of the functions shown below? <code>float('-infinity') float('inf')</code>	-inf inf
What is the output shape of the code shown? <code>import turtle t=turtle.Pen() for i in range(0,4): t.forward(100) t.left(120)</code>	triangle
Is the following code valid? <code>try: # Do something except: # Do something else: # Do something</code>	yes
What is the output of the following? <code>try: if '1' != 1: raise</code>	invalid code

<code>"someError" else: print("someError has not occurred") except "someError": print ("someError has occurred")</code>	
Which of the following statements is true?	The standard exceptions are automatically imported into Python programs.
What is the biggest reason for the use of polymorphism?	The program will have a more elegant design, and will be easier to maintain and update
What is the output of the following piece of code? <code>class A: def __repr__(self): return "1" class B(A): def __repr__(self): return "2" class C(B): def __repr__(self): return "3" o1 = A() o2 = B() o3 = C() print(obj1, obj2, obj3)</code>	1 2 3

Question	Correct Answer
<code>r = lambda q: q * 2 s = lambda q: q * 3 x = 2 x = r(x) x = s(x) x = r(x) print x</code>	24
<code>a = 4.5 b = 2 print a//b</code>	2.0
<code>a = True b = False c = False if a or b and c: print "PYTHON IS BEST" else: print "R IS BEST "</code>	PYTHON IS BEST
<code>a = True b = False c = False if not a or b: print 1 elif not a or not b and c: print 2 elif not a or b or not b and a: print 3 else: print 4</code>	3
<code>count = 1 def doThis(): global count for i in (1, 2, 3): count += 1 doThis() print count</code>	4
<code>class Acc: def __init__(self, id): self.id = id id = 555 acc = Acc(111) print acc.id</code>	111
<code>for i in range(2): print i for i in range(4,6): print i</code>	0 1 4 5
<code>values = [1, 2, 3, 4] numbers = set(values) def checknums(num): if num in numbers: return True else: return False for i in filter(checknums, values): print i</code>	1 2 3 4
<code>counter = {} def addToCounter(country): if country in</code>	3

counter: counter[country] += 1 else: counter[country] = 1 addToCounter('China') addToCounter('Japan') addToCounter('china') print len(counter)	
class Geeks: def __init__(self, id): self.id = id manager = Geeks(100) manager.__dict__['life'] = 49 print manager.life + len(manager.__dict__)	51
a = "PYTHONISBEST" b = 13 print a + b	AN ERROR SHOWN
dictionary = {} dictionary[1] = 1 dictionary['1'] = 2 dictionary[1] += 1 sum = 0 for k in dictionary: sum += dictionary[k] print sum	4
dictionary = {1:'1', 2:'2', 3:'3'} del dictionary[1] dictionary[1] = '10' del dictionary[2] print len(dictionary)	2
nameList = ['Harsh', 'Pratik', 'Bob', 'Dhruv'] print nameList[1][-1]	k
nameList = ['Harsh', 'Pratik', 'Bob', 'Dhruv'] pos = nameList.index("PYTHON") print pos * 5	An Exception is thrown, ValueError: 'PYTHON' is not in list
geekCodes = [1, 2, 3, 4] geekCodes.append([5,6,7,8]) print len(geekCodes)	5
def addToList(listcontainer): listcontainer += [10] mylistContainer = [10, 20, 30, 40] addToList(mylistContainer) print len(mylistContainer)	5
def gfgFunction(): "PYTHON is cool language for boosting up technical skills" return 1 print gfgFunction.__doc__[10:24]	cool
class A(object): val = 1 class B(A): pass class C(A): pass print A.val, B.val, C.val B.val = 2 print A.val, B.val, C.val A.val = 3 print A.val, B.val, C.val	1 1 1 1 2 1 3 2 3
check1 = ['Learn', 'Quiz', 'Practice', 'Contribute'] check2 = check1 check3 = check1[:] check2[0] = 'Code' check3[1] = 'Mcq' count = 0 for c in (check1, check2, check3): if c[0] == 'Code': count += 1 if c[1] == 'Mcq': count += 10 print count	12
def gfg(x,l=[]): for i in range(x): l.append(i*i) print(l) gfg(2) gfg(3,[3,2,1]) gfg(3)	[0, 1] [3, 2, 1, 0, 1, 4] [0, 1, 0, 1, 4]

<pre>#code for disabling the softspace feature print('G','F','G', sep='') #for formatting a date print('09','12','2016', sep='-') #another example print('pratik','geeksforgeeks', sep='@')</pre>	GFG 09-12-2016 pratik@geeksforgeeks
<pre># Python code to demonstrate Type conversion # using int(), float() # initializing string s = "10010" # printing string converting to int base 2 c = int(s,2) print ("After converting to integer base 2 : ", end="") print (c) # printing string converting to float e = float(s) print ("After converting to float : ", end="") print (e)</pre>	After converting to integer base 2 : 18 After converting to float : 10010.0
<pre># Python code to demonstrate Type conversion # using ord(), hex(), oct() # initializing integer s = '4' # printing character converting to integer c = ord(s) print ("After converting character to integer : ",end="") print (c) # printing integer converting to hexadecimal string c = hex(56) print ("After converting 56 to hexadecimal string : ",end="") print (c) # printing integer converting to octal string c = oct(56) print ("After converting 56 to octal string : ",end="") print (c)</pre>	After converting character to integer : 52 After converting 56 to hexadecimal string : 0x38 After converting 56 to octal string : 0o70
<pre>var1 = 'Python is best ' var2 = "Python is ultimate" print "var1[0]: ", var1[0] # statement 1 print "var2[1:5]: ", var2[1:5] # statement 2</pre>	var1[0]: P var2[1:5]: ytho
<pre>var1 = 'Python' print "Original String :-", var1 print "Updated String :- ", var1[:5] + 'for' + 'IOT' # statement 1</pre>	Original String :- Python Updated String :- PythonforIOT
<pre>list = [1, 2, 3, None, (1, 2, 3, 4, 5), ['Python', 'for', 'IOT']] print len(list)</pre>	6
<pre>list = ['a', 'b', 'c', 'd', 'e'] print list[10:]</pre>	[]
<pre>list = ['a', 'b', 'c']*3 print list</pre>	[]
<pre># importing "collections" for deque operations import collections # initializing deque de = collections.deque([1, 2, 3, 3, 4, 2, 4]) # using index() to print the first occurrence of 4 print ("The number 4 first occurs at a position : ") print (de.index(4,2,5))</pre>	The number 4 first occurs at a position : 4

Question	Correct Answer
Which of the following statements best describes a white-hat hacker?	Security professional
A security audit performed on the internal network of an organization by the network administration is also known as _____.	White-box testing
What is the first phase of hacking?	Reconnaissance
Which type of hacker represents the highest risk to your network?	Disgruntled employees
Which are the four regional Internet registries?	APNIC, LACNIC, ARIN, RIPE NCC
_____ framework made cracking of vulnerabilities easy like point and click.	Metasploit
_____ is a popular tool used for discovering networks as well as in security auditing.	Nmap
Which of this Nmap do not check?	what type of antivirus is in use
Wireshark is a _____ tool	network protocol analysis
Which of the below-mentioned tool is used for Wi-Fi hacking?	Aircrack-ng
Performing a shoulder surfing in order to check other's password is _____ ethical practice.	a bad
_____ has now evolved to be one of the most popular automated tools for unethical hacking.	Malware
A penetration tester must identify and keep in mind the _____ & _____ requirements of a firm while evaluating the security postures.	privacy and security
The attacker will use different bots (zombie PCs) to ping your system and the name of the attack is _____	Distributed Denial-of-Service (DDoS)
A _____ takes over your system's browser settings and the attack will redirect the websites you visit some other websites of its preference.	Browser Hijacker
_____ is a malicious method used by cyber-	Clickjacking

criminals to trick a user into clicking on something different from what the user wants.	
What are the forms of password cracking techniques?	All of the above
What is the attack called "evil twin"?	Rogue access point
_____ is a component of the reconnaissance stage that is used to gather possible information for a target computer system or network.	Footprinting
How many types of footprinting are there?	2
_____ is one of the 3 pre-attacking phase.	Footprinting
Which of them is not an information source over the internet for target attackers?	YouTube
Spywares can be used to steal _____ from the attacker's browser.	browsing history
Ping sweep is also known as _____	ICMP sweep
_____ is a common tool used for doing OS fingerprinting.	Nmap
_____ is an ethical hacking technique used for determining what operating system (OS) is running on a remote computer.	OS fingerprinting
XSS is abbreviated as _____	Cross Site Scripting
When there is an excessive amount of data flow, which the system cannot handle, _____ attack takes place.	Buffer Overflow attack
Which of them is not a wireless attack?	Phishing
Which method of hacking will record all your keystrokes?	Keylogging